

JIMMY LIU

@ jimmy.liu@uwaterloo.ca

ji-mmyliu

jimmyliu.dev

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647-567-9466

Toronto, ON

EXPERIENCE

Software Engineer Intern @ Aven

May 2026 – August 2026 San Jose, CA

- Built and shipped end-to-end features across Vue.js, Python, Node.js, TypeORM, and PostgreSQL, increasing balance transfer conversion from 8% → 30% and Home Equity loan upgrade conversion from 1% → 5%; balance transfer flow drove ~\$10K in weekly transferred balances.
- Designed A/B experiment analytics and SQL dashboards in Snowflake to measure product performance, partnering with product managers, designers, data scientists, and credit analysts to iterate on user-facing features.
- Processed Experian credit data and implemented hard-cut rules to tighten the credit box and reduce delinquency
- Strengthened reliability by writing and debugging Playwright E2E tests, including resolving flaky test failures.

Software Engineer Intern @ Shopify

September 2025 – December 2025 Toronto, ON

- Built developer shortcuts to automatically compile the Shopify POS mobile app for end-to-end testing, integrating with Appium through Ruby and Bash scripting. Eliminated flakiness in critical E2E tests, enabling merge-queue adoption.
- Simplified Bitrise CI debugging by building an AI log analyzer to surface root causes from verbose Appium failure logs.

Software Engineer Intern @ Spellbook

January 2025 – April 2025 Toronto, ON

- Boosted weekly active users by 12% through development of advanced legal playbook configurations via React.js Microsoft Word extension and Express backend.
- Deployed Spellbook's freemium site through the AWS CDK, patched cybersecurity vulnerabilities, enabling potential clients to trial key features before purchasing, driving a 14% increase in user conversions.

Lead Software Engineer @ UW Blueprint

May 2024 – Present Waterloo, ON

- Led engineering team in developing a React.js web application for non-profit organization Extend-A-Family, enabling delivery of financial literacy lessons for children with developmental disabilities.
- Drove system design and communication between product managers and the non-profit organization.

PROJECTS

Screentime Showdown devpost.com/software/screentime-showdown

- Competition site that helps people overcome social media addiction, made with React.js, Express, and Instagram API.
- Won Hack the Valley 2023's Best Financial Hack Award.

Laser Guitar github.com/ji-mmyliu/laser-guitar

- Designed and built an electric guitar with lasers for strings!
- Python, Raspberry Pi, and self-trained Tensorflow Keras ML model for chord fingering detection.

Wireguard VPN Wizard github.com/ji-mmyliu/wg-vpn-onboarder

- CLI that configures and deploys custom virtual private networks utilizing Wireguard technology, written in Golang.

Discord Bot Online Judge github.com/DiscordBotOnlineJudge

- An open-source public competitive programming online judge platform hosting 10 contests, 73 total puzzles, 179 users worldwide, and support for 18 programming languages.
- Developed a Discord Application using Python, MongoDB, gRPC, and an additional Flask web frontend.

TECHNICAL KNOWLEDGE

- Programming Languages:** Python, Java, C, C++, Go, Ruby, Typescript, Bash, Scala
- Frameworks:** React.js, Vue.js, Express.js, Flask, Django, Tensorflow, AWS, Linux, Git, Docker, Kubernetes, Jenkins, Snowflake, Grafana, Claude Code

AWARDS

- Competitive Programming:** Canadian Computing Competition Senior Division – top 5%; USACO – Gold Division
- Cybersecurity:** Carnegie Mellon University picoCTF – 3rd place in Canada

EDUCATION

University of Waterloo

Bachelor of Software Engineering / Quantum Engineering Option (92% Average)

September 2023 – April 2028